Type MS · Find and fix errors in an algorithm

Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.iii in this mark scheme extract

6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Find and fix errors in an algorithm

Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.iii in this mark scheme extract

6(b)(iv)	Award **one** mark for: • On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1) `LENGTH(arrival) - (late + early)`	Allow number of trains to be hard coded as 7 in place of `LENGTH(arrival)` Allow equivalent expressions: `LENGTH(arrival) - late - early`	1
6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Find and fix errors in an algorithm

Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.iii in this mark scheme extract

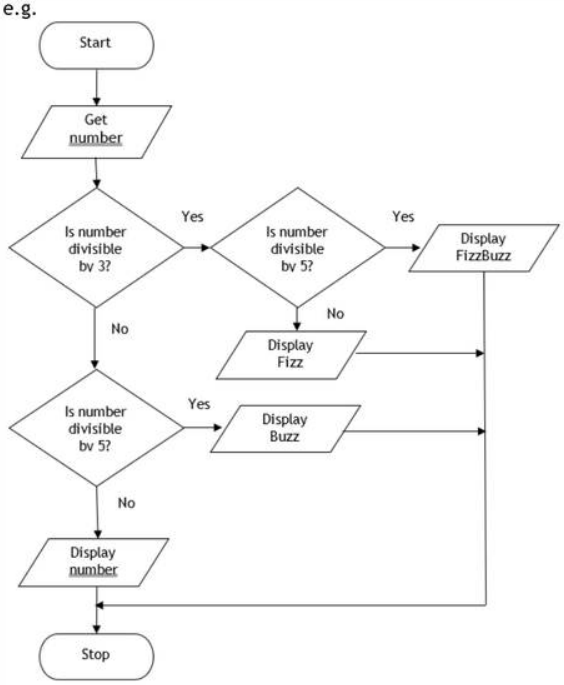
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Find and fix errors in an algorithm

Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.iii in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Find and fix errors in an algorithm

Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.iii in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Find and fix errors in an algorithm

Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

5(a)(i)	One mark for each correct row **Width** / **Length** / **Height** / **Output** 4 / 4 / 2 / Requires large container (1) 2 / 2 / 2 / Too small for cargo container (1) 3 / 8 / 5 / Too large for cargo container (1) Do not penalise spelling	3

Question Number	Answer	Additional Guidance	Mark
5(a)(ii)	Any response that indicates the concept of: Determining which size shipping container is needed for the indicated cargo.		1

Type MS · Find and fix errors in an algorithm

Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)	SEND states [numStates] TO DISPLAY (1) or SEND state TO DISPLAY (1) SET numStates TO numStates + 1 (1) or LENGTH(states) (1) Concatenation of message and variable uses " and & & numStates & TO DISPLAY (1) or & LENGTH(states) & TO DISPLAY (1) or STRING(numStates) TO DISPLAY (1) Pseudocode is correct (1)	• Accept , or + in place of & • Ignore capitalisation of commands •	4

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

Type MS · Find and fix errors in an algorithm

Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Showing that index goes up to 4 and does not exceed 4 (1) Showing correspondence between values of index and loadWeight[index] (1)	• There does not have to be an explicit indication of the out-of-bounds / crashing error. • Ignore target and found columns • Allow two rows per loop with matching index and loadWeight values	2

target	found	index	loadWeight[index]
50500	FALSE	0	20000
		1	28000
		2	40000
		3	50000
		4	

target	found	index	loadWeight[index]
50500	FALSE	0	
		1	20000
		2	28000
		3	40000
		4	50000

Question Number	Answer	Additional Guidance	Mark
5(c)(ii)	1. Use of AND operator (in the test of relevant conditions) (1) 2. Use of suitable function to find the length of the array using a correct comparison to index (1) Example: WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO	• Do not penalise pseudocode syntax • Ignore changes to first condition as long as meaning is not lost	2

Type MS · Find and fix errors in an algorithm

Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td></td><td></td><td>10</td><td>3</td><td></td><td></td><td></td><td>10</td></tr>
<tr><td>2</td><td></td><td></td><td>6</td><td></td><td>9</td><td></td><td>6</td><td></td></tr>
</table>

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td>4</td><td>2</td><td>10</td><td>3</td><td>6</td><td>1</td><td>9</td><td>10</td></tr>
<tr><td>2</td><td>4</td><td>2</td><td>6</td><td>3</td><td>9</td><td>1</td><td>6</td><td>10</td></tr>
</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Find and fix errors in an algorithm

Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Find and fix errors in an algorithm

Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td></td><td></td><td>10</td><td>3</td><td></td><td></td><td></td><td>10</td></tr>
<tr><td>2</td><td></td><td></td><td>6</td><td></td><td>9</td><td></td><td>6</td><td></td></tr>
</table>

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td>4</td><td>2</td><td>10</td><td>3</td><td>6</td><td>1</td><td>9</td><td>10</td></tr>
<tr><td>2</td><td>4</td><td>2</td><td>6</td><td>3</td><td>9</td><td>1</td><td>6</td><td>10</td></tr>
</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Find and fix errors in an algorithm

Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Find and fix errors in an algorithm

Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 5 of 12 · Question 5 part b.iii · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part b.iii in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td></td><td></td><td>10</td><td>3</td><td></td><td></td><td></td><td>10</td></tr>
<tr><td>2</td><td></td><td></td><td>6</td><td></td><td>9</td><td></td><td>6</td><td></td></tr>
</table>

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td>4</td><td>2</td><td>10</td><td>3</td><td>6</td><td>1</td><td>9</td><td>10</td></tr>
<tr><td>2</td><td>4</td><td>2</td><td>6</td><td>3</td><td>9</td><td>1</td><td>6</td><td>10</td></tr>
</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Find and fix errors in an algorithm

Answer 5 of 12 · Question 5 part b.iii · 1 mark

2021 JUN · Paper 01 · extract 2 of 3 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Find and fix errors in an algorithm

Answer 5 of 12 · Question 5 part b.iii · 1 mark

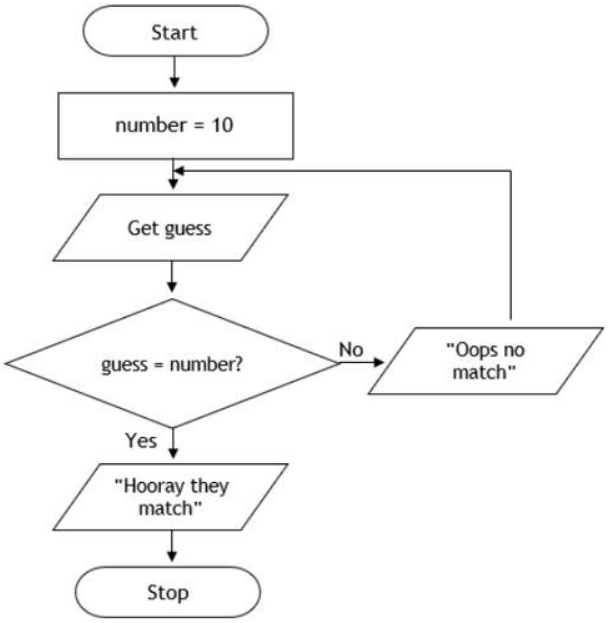
2021 JUN · Paper 01 · extract 3 of 3 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Find and fix errors in an algorithm

Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Find and fix errors in an algorithm

Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part c.ii in this mark scheme extract

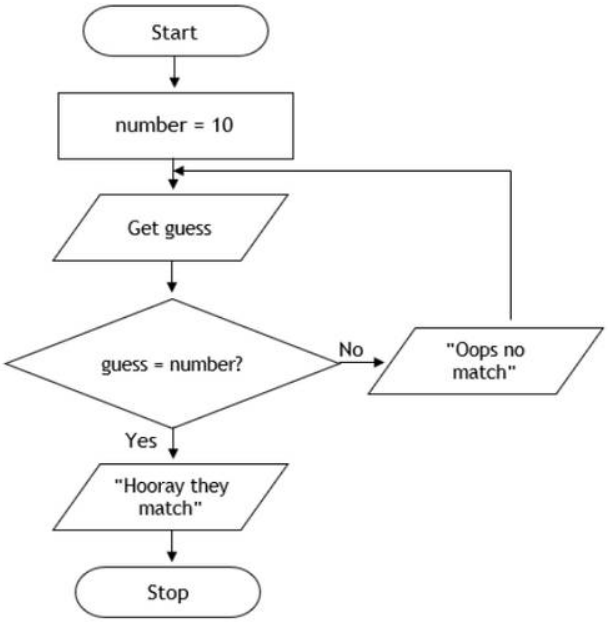
4(c)(iii)	Award **one** mark for: • Pseudododode that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part c.iii in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part c.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part c.iii in this mark scheme extract

4(c)(iii)	Award **one** mark for: • Pseudododode that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 1 of 9 · find part b in this mark scheme extract

5(a)	A1 A2	Award 1 mark for demonstration of one correctly merged/split row Award 2 marks for correct merge sort Blackfin, Bigeye | Longtail, Albacore | Bluefin Longtail, Blackfin, Bigeye, Albacore | Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Bigeye | Blackfin, Albacore | Longtail, Bluefin Bigeye | Longtail, Bluefin, Blackfin, Albacore Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye | Longtail, Bluefin, Albacore	We are testing pupils understanding of how a merge sort progresses: • Divide and conquer (split) • Merging adjacent elements Allow answers in ascending order	2

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 2 of 9 · find part b in this mark scheme extract

		Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye, Albacore | Longtail, Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore		

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 3 of 9 · find part b in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(b)</td><td>B1
B2</td><td>Award one mark for each correct cell:
<table><tr><td>Line number with error</td><td>11 (1)</td></tr><tr><td>Corrected line of pseudocode</td><td>SET tmp TO myTuna[ndx] (1)</td></tr></table></td><td>Ignore spellings

Ignore missing text as long as the minimum of [ndx] is provided

B2 does not depend on B1</td><td>2</td></tr>
</table>

Question	mp	Answer	Additional guidance	Mark
5(c)	Award **one** mark for each of:		C1 allow append as writing to file C4 Do not award comma within an array C5 must have 1 complete line written to file (allow additional spaces and commas). Should include individual elements (2D index) and not a 1D array from tbl_tuna C6 complete file must be as expected e.g. no additional comma at end of line and no spaces around commas	**6**

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 4 of 9 · find part b in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String[,] tblTuna =  {
    { "Yellowfin", "105", "15", "3"},
    { "Albacore","90","15","5"},
    { "Skipjack","50","3","4"},
    { "Bigeye","105","25","4"},
    { "Atlantic Bonito","50","4","2"},
    { "Northern Bluefin","190","120","11"},
    { "Southern Bluefin","190","120","11"},
    { "Tongol","90","20","4"}
};



// ------------------------------------------
// Write your code below this line

String fileName = "TunaData.txt";       // Output file name
String comma = ",";                     // Use as a constant
String full_path = "C:\\Q05c";
fileName = full_path + "\\" + fileName;

// Create variables as needed
int number = 101;           // Number for the line at the front
String lineOut = "";        // The line to output

// Get the dimensions of the array for looping
int rows = tblTuna.GetLength(0);
int columns = tblTuna.GetLength(1);

// Open the file for writing
StreamWriter fileWriter = new StreamWriter(fileName);

```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 5 of 9 · find part b in this mark scheme extract

```
37  // Process every tuna in the table
38  for (int i = 0; i < rows; i++)          // Rows
39  {
40      // Start with the number and a comma
41      lineOut = number.ToString() + comma;
42
43      for (int j = 0; j < columns; j++)   // Columns
44      {
45          // Add the name and the numbers
46          lineOut = lineOut + tblTuna[i, j];
47
48          // Add a comma to all the fields except the last
49          if (j < columns - 1)
50          {
51              lineOut = lineOut + comma;
52          }
53      }
54      // Write the line to the file
55      fileWriter.WriteLine(lineOut);
56
57      // Go to the next number for the line
58      number = number + 1;
59  }
60
61  // Close the file
62  fileWriter.Close();
```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 6 of 9 · find part b in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileWriter;
import java.io.IOException;

public class Q05cFINISHED {

    public static void main(String[] args) throws Exception {

        String[][] tblTuna =
        {
            { "Yellowfin", "105", "15", "3"},
            { "Albacore","90","15","5"},
            { "Skipjack","50","3","4"},
            { "Bigeye","105","25","4"},
            { "Atlantic Bonito","50","4","2"},
            { "Northern Bluefin","190","120","11"},
            { "Southern Bluefin","190","120","11"},
            { "Tongol","90","20","4"}
        };

        // ------------------------------------------
        // Write your code below this
        String fileName = "TunaData.txt";       // Output file name
        String comma = ",";                     // Use as a constant
        String full_path = "C:\\src\\q05c";
        fileName = full_path + "\\" + fileName;

        // Create variables as needed
        int number = 101;           // Number for the line at the front
        String lineOut = "";        // The line to output
        String linefeed = "\n";     // Use as a constant

        // Get the dimensions of the array for looping
        int rows = tblTuna.length;
        int columns = tblTuna[0].length;

        // Open the file for writing
        FileWriter fileWriter = new FileWriter(fileName);

```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 7 of 9 · find part b in this mark scheme extract

```
41          // Process every tuna in the table
42          for (int i = 0; i < rows; i++)          // Rows
43          {
44              // Start with the number and a comma
45              lineOut = Integer.toString(number) + comma;
46
47              for (int j = 0; j < columns; j++)   // Columns
48              {
49                  // Add the name and the numbers
50                  lineOut = lineOut + tblTuna[i][j];
51
52                  // Add a comma to all the fields except the last
53                  if (j < columns - 1)
54                  {
55                      lineOut = lineOut + comma;
56                  }
57                  else
58                  {
59                      lineOut = lineOut + linefeed;
60                  }
61              }
62              // Write the line to the file
63              fileWriter.write(lineOut);
64
65              // Go to the next number for the line
66              number = number + 1;
67          }
68
69          // Close the file
70          fileWriter.close();
71      }
72
73  }
```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 8 of 9 · find part b in this mark scheme extract

Python

```
# Q05cFINISHED

tbl_tuna = [["Yellowfin",105,15,3],
            ["Albacore",90,15,5],
            ["Skipjack",50,3,4],
            ["Bigeye",105,25,4],
            ["Atlantic Bonito",50,4,2],
            ["Northern Bluefin",190,120,11],
            ["Southern Bluefin",190,120,11],
            ["Tongol",90,20,4]]

# ---------------------------------------------
# Write your code below this line

FILE_OUT = "TunaData.txt"           # Output file name
COMMA = ","                         # Use as constant

# Create variables as needed
LINEFEED = "\n"
number = 101                        # Number at the front
line_out = ""                       # The line to write

# Open the file for writing
file = open (FILE_OUT, "w")

# Process every tuna in the table
for tuna in tbl_tuna:
    # Start with the number and a comma
    line_out = str (number) + COMMA

    # Add the name of the tuna and a comma
    line_out = line_out + tuna[0] + COMMA

    # Each of the numbers need to be converted to a string
    # before adding to the output string.
    line_out = line_out + str (tuna[1]) + COMMA
    line_out = line_out + str (tuna[2]) + COMMA
    line_out = line_out + str (tuna[3])     # No comma on last field

    # Add a line feed to the whole line
    line_out = line_out + LINEFEED

    # Write the line to the file
    file.write (line_out)

    # Go to the next number for the line
    number = number + 1

# Close the file
file.close ()
```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 9 of 9 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Find and fix errors in an algorithm

Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

Number			
6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Find and fix errors in an algorithm

Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Type MS · Find and fix errors in an algorithm

Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

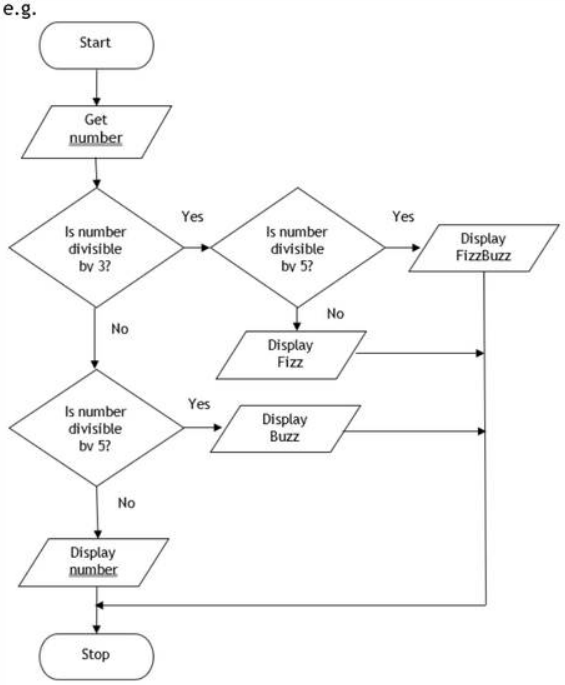
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Find and fix errors in an algorithm

Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

22

Type MS · Find and fix errors in an algorithm

Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.i in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Find and fix errors in an algorithm

Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Find and fix errors in an algorithm

Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Type MS · Find and fix errors in an algorithm

Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

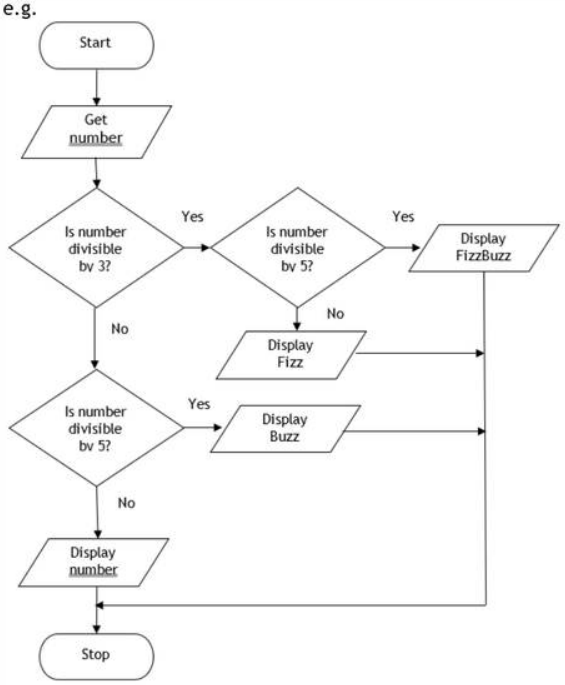
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Find and fix errors in an algorithm

Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Find and fix errors in an algorithm

Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Find and fix errors in an algorithm

Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.iv in this mark scheme extract

6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Find and fix errors in an algorithm

Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.iv in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Find and fix errors in an algorithm

Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.iv in this mark scheme extract

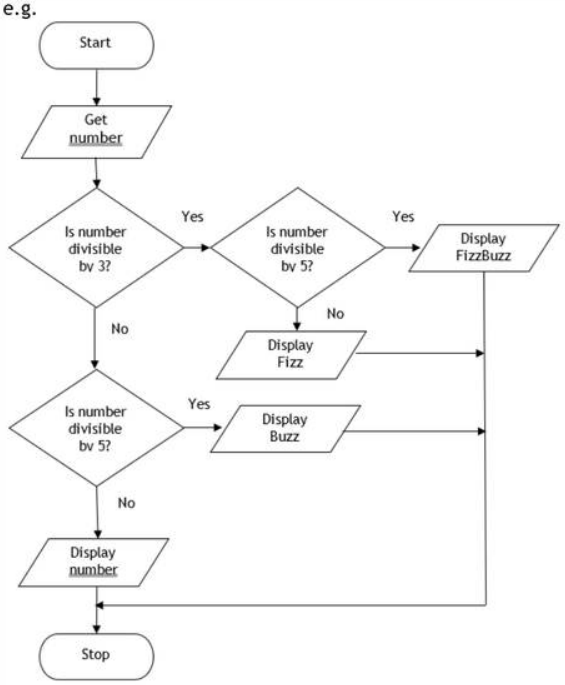
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Find and fix errors in an algorithm

Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.iv in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Find and fix errors in an algorithm

Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.iv in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Find and fix errors in an algorithm

Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part a.i in this mark scheme extract

5(a)(i)	A1 A2	final score not accurate (1) because score is resetting/score keeps going back to zero (in each iteration of the loop) (1)	'Score not accurate' is not enough on its own 'Score on line 20' or 'Score output to DISPLAY' would count as final score.	**2**

Question	mp	Answer	Additional guidance	Mark
5(a)(ii)	A3 A4	Award up to **two** marks for a linked description such as: • Asking 10 addition questions (1) and calculating final score (1) • Asking randomised adding questions (1) and checking user answer against computer answer (1) • Creating adding quiz with random numbers up to 99 (1) and outputting number of correct answers (1)	Processes that can be included Concept of this being an 'Adding Quiz' required • Getting random numbers for question • Calculating sum of two random numbers for correct answers • comparing user answer to correct answer • incrementing /tracking score • asking 10 questions (loop) • outputting final score	**2**

Type MS · Find and fix errors in an algorithm

Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
5(b)				
	B1	Open file for writing (1)	"w" mode only Accept "w+"	
	B2	Selection used to get grade (1)	Must address all 4 grades Does not have to be accurate on boundaries e.g. >80 is ok for distinction etc.1	
	B3	Output suitable message to say file created (1)		
	B4	Evidence of 2D indexing (1)	Even if element not used in rest of program 2D indexing **or** 1D index within *for-each* structure **or** sum(1Darray) or min(1Darray) etc. At least 1 inner value accessed	
	B5	Lowest score **must** be found and omitted/subtracted from calculated scores (all arrays) (1)	Any method e.g.: • List methods • Selection statements • iterable functions must be done programmatically	
	B6	At least 1 line written to file (1)	Must contain something generated by program (total/result), does not have to be correct total or grade	
	B7	Add line feed to write line after both score and grade (1)		
	B8	Output file all correct (1)		**8**

18 of 23

Type MS · Find and fix errors in an algorithm

Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part a.i in this mark scheme extract


```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

with open ("grades.txt","w") as theFile:

    # calculate each students score
    for student in Tbl_scores:
        student.sort()
        total = student[1] + student[2]

        # decide student grade
        if total >= 80:
            grade = "Distinction"
        elif total >= 65:
            grade = "Merit"
        elif total >= 50:
            grade = "Pass"
        else:
            grade = "Fail"

        # write score and grade to file
        outString = str(total) + " " + grade + "\n"
        theFile.write(outString)

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = sum(student) - min(student)

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = 0
    lowest = 51
    for score in student:
        if score < lowest:
            lowest = score
        total = total + score
    total = total - lowest

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

Type MS · Find and fix errors in an algorithm

Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark